Emmanuel Idoko

eemmanuel.idoko@gmail.com | linkedin.com/in/emmanuelidoko | github.com/pidoxy

EDUCATION

University of Lagos — Lagos, Nigeria
Bachelor of Science in Computer Science; CGPA: 4.68/5.0 — *Expected 2027*

- **Relevant Coursework:** Discrete Mathematics, Data Structures, Computational Methods, Linear Algebra, Calculus, Python/C Programming, Net Centric Computing, Algorithm and Complexity, Compiler Construction, Concurrent Programming, Operations Research, Systems Analysis, Operating Systems, Machine Assembly Language.

African Computer Vision Summer School (ACVSS) — Accra, Ghana
3rd ACVSS, Google AI Community Center — *Jul 2026*

- **Sessions:** Self-supervised learning, multimodal learning and vision-language models, generative and diffusion models, 3D computer vision, fairness and responsible computer vision, nature-inspired approaches to multimodal vision, and generative-model inference and evaluation; plus keynotes, research talks, poster sessions, and a hackathon.
- **Hackathon & poster:** Built a task-aware super-resolution pipeline for satellite flood mapping in the research hackathon; presented the SharpXR poster at the school's poster session.

Udacity — Remote
AI Programming with Python Nanodegree — *Oct 2022 – Jan 2024*

PUBLICATIONS & ONGOING PROJECTS

Retrieval-Induced Hallucination in Medical Vision-Language Models
Submitted — Africa in AI Workshop, NeurIPS 2026 (Springer LNCS) | First author

- Found that 95% of retrieval-augmented chest X-ray reports copy text verbatim from another patient's report, against 0% without retrieval, across 100 MIMIC-CXR studies.
- Showed that retrieval doubles clinical accuracy when the retrieved case is relevant (CheXbert F1 0.201 to 0.402) but collapses it to 0.043 when mismatched.

VAMAE: Vessel-Aware Masked Autoencoders for OCT-Angiography
Accepted — 28th International Conference on Pattern Recognition (ICPR 2026) | arXiv:2604.06583

- **Objective:** Develop a self-supervised pre-training method for OCT-A image analysis that learns robust representations of retinal vasculature.
- **Methodology:** Proposing a masked autoencoder (MAE) variant with a vessel-aware loss function that prioritizes reconstruction of fine vascular structures to improve downstream segmentation and classification.

SharpXR: Structure-Aware Denoising for Pediatric Chest X-Rays
Published — MIRASOL Workshop, MICCAI 2025 | pp. 83–92 | arXiv:2508.08518

- Contributed to the experimental evaluation of a structure-preserving denoising method for noisy pediatric chest X-rays, benchmarking seven baseline models (REDCNN, DnCNN, HFormer, ResUNet++, Attention U-Net, Sharp U-Net, BM3D).
- Demonstrated improved downstream pneumonia classification accuracy from 88.8% to 92.5% using SharpXR-denoised images.
- Wrote the abstract and introduction; performed proofreading and editing of the full manuscript.

Research Poster Presentation
Machine Intelligence Research Group International Conference (MIRG-ICAIR 2025)

- Presented a research poster on SharpXR at MIRG-ICAIR 2025.

RESEARCH EXPERIENCE

AidCare — Lagos, Nigeria (Hybrid)
AI Research Lead & Co-Founder — *May 2025 – Present*

- Architected a dual-mode Retrieval-Augmented Generation (RAG) system to provide context-aware clinical decision support, leveraging distinct knowledge bases for physicians and community health workers.
- Engineered a semantic search pipeline using Sentence Transformers and FAISS to index and query a heterogeneous corpus of medical textbooks and Nigerian national guidelines.
- Developed a multi-stage LLM prompting framework (Google Gemini) for grounded reasoning, enabling the synthesis of patient data with retrieved knowledge to generate differential diagnoses.
- Designed a multi-modal data ingestion pipeline integrating local Whisper AI for transcription and Tesseract OCR for patient document processing.

ML Collective — Remote
Independent ML Researcher — *Sep 2024 – Present*

- Engaged in weekly critical analysis and presentation of seminal and contemporary research papers in machine learning, with a focus on medical imaging and clinical NLP.
- Collaborated within focus groups to identify open research questions, formulate novel hypotheses, and design experimental protocols for computer vision research projects.

Wema Bank Lagos, Nigeria (Hybrid)
Data and AI Internship *Sep 2024 – Dec 2024*

- Researched and built a governance-compliant pipeline to extract and analyze Play Store reviews, producing actionable insights for product teams.
- Contributed to the development of an AI-powered knowledge chatbot for Microsoft Teams, enabling 2,000+ staff to instantly access product and process information, reducing support query time.

ENGINEERING EXPERIENCE

HabariPay – GTCO Lagos, Nigeria (Hybrid)
Software Engineer, Contract *Sep 2024 – Present*

- Built a full-stack vendor guide app (Next.js, Node.js, PostgreSQL) for a food festival, used by 260+ attendees and generating 2,100+ page views.
- Automated ingestion of vendor data from 2 sources into a centralized SQLite DB, processing hundreds of records with high consistency.
- Implemented a TF-IDF-based recommendation engine and Gemini-powered summaries, improving user click-through rate by 7%.

NexaScale Remote
Cloud Engineer Internship *Feb 2023 – May 2023*

- Automated deployment of 30+ AWS resources with Terraform, reducing manual setup by 75%.
- Developed a microservice-based application with Node.js, containerized with Docker, and deployed to production with Kubernetes (AKS).
- Streamlined deployments with a CI/CD pipeline (Azure DevOps, Git, Docker) for fast and reliable releases.

ZuriChat Remote
Software Engineer & Lead Internship *Jul 2021 – Oct 2021*

- Led a team to ship real-time music and chess rooms using WebRTC, boosting user engagement by 20%.
- Improved backend performance by 50% through caching, code refactoring, and API optimization.
- Managed multi-team deployments with GitHub Actions and Jenkins, ensuring stable integrations across features.

PROJECTS

Virtue Foundation Intelligence Platform Python, LangGraph, FAISS, GPT-4o-mini, Next.js

- Built an end-to-end agentic AI platform to detect medical deserts and analyze healthcare gaps across 797 Ghana facilities in all 16 regions.
- Designed an Intelligent Document Processing (IDP) pipeline using GPT-4o-mini to extract structured capability data from unstructured facility records, with per-field confidence scoring (0-99%).
- Architected a LangGraph supervisor multi-agent system classifying natural language queries into 6 categories and routing to specialized sub-agents backed by a FAISS vector store on OpenAI embeddings.
- Identified 10 medical deserts across 16 regions, flagged 43 data anomalies across 39 facilities, and generated AI-powered deployment recommendations; integrated ElevenLabs voice I/O for field accessibility.
- Won 2nd place at Databricks x Hack-Nation Global AI Hackathon.

Kinetix Python, FastAPI, Next.js, TypeScript, Gemini API, WebSockets

- Built a real-time AI personal trainer using Gemini's vision capabilities to analyze webcam exercise footage frame-by-frame, classifying posture and joint alignment as safe, suboptimal, or unsafe.
- Designed a dual-persona prompt architecture: an internal Analyst agent performs biomechanical reasoning while an external Coach agent delivers concise real-time verbal coaching via text-to-speech.
- Implemented WebSocket communication between FastAPI backend and Next.js frontend for low-latency feedback, with skeleton overlay and color-coded form indicators (green/yellow/red).
- Generates post-session summaries with rep counts, recurring problem areas, and corrective exercise recommendations.

PheraCAM Python, React.js, Azure, OpenAI API

- Developed a real-time face recognition system leveraging OpenCV and Azure cloud services for scalable, low-latency processing.
- Achieved >95% recognition accuracy with <150ms response time.
- Won 2nd place among 50+ teams at Hacklab's Hackathon, recognized for technical innovation and robustness.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL, C, Bash
- **ML & Imaging:** PyTorch, Hugging Face (Transformers, Sentence Transformers), Scikit-learn, FAISS, Pandas, NumPy

- **Frameworks/Tools:** FastAPI, Flask, Node.js, Next.js, Docker, PostgreSQL, MongoDB, DigitalOcean, Git, Linux

AWARDS & COMMUNITY LEADERSHIP

Project Lead | Unilag Data Community Dec 2024 – Present

- Developed and delivered a comprehensive workshop series (Python, SQL, Power BI, Excel), successfully training over 250 students.
- Mentored beginner and intermediate members on project scoping, methodology, and technical implementation.

Deputy Trainings & Competitions Lead | GDSC UNILAG

- Coordinated internal hackathons and competitions, fostering innovation and practical problem-solving.
- Designed and led technical training sessions and hands-on workshops.

SCHOLARSHIPS

- Jim Ovia Undergraduate Scholarship (2025–2027): National scholarship recognizing top performance in STEM.
- Squad Hackademy Undergraduate Scholarship (2024–2027): Recognizing top performance in programming.
- PICFI Undergraduate Scholarship (2025–2027): National scholarship recognizing top performance in STEM.